HED 8.0 RE:IMAGINE • ROUND 1 / ROUND 2 PLANNING

WASTEWISE VERIFY

Verify Before Dispatch

Revised Master Project Brief & Execution Plan

TRACK	CHALLENGE	PROBLEM STATEMENT
RE:MAKE	HED8-MAK-04	Waste as a Resource

PRIMARY HERO PATHWAY	TEAM COMPOSITION	CORE DIFFERENTIATOR
Source-segregated C&D rubble → screened aggregate → registered recycler → paving blocks	5 members: 2 Mechatronics + 3 AI/ML	Physical verification + QR batch passport + compatibility-first pooling + delivered-cost ranking

1. Executive Thesis

Construction and demolition waste is not always landfilled because buyers do not exist. A major barrier is the uncertainty buyers face before paying to transport and process mixed, small or unverified loads. Photographs and supplier listings cannot establish the usable mass, visible contamination, processing loss or delivered cost of a specific batch.

> **THE CORE ASSUMPTION WE CHALLENGE**
>
> Industrial waste exchange needs more than another online listing marketplace. Buyers need credible pre-dispatch evidence: what was received, what was screened, how much usable mineral material was recovered, and what the material is likely to cost after processing and transport.

2. WasteWise Verify in One Paragraph

WasteWise Verify proposes a shared screening station at an appropriately authorised C&D collection or intermediate-storage facility. It sieves and presents construction rubble in a controlled layer, uses RGB vision to identify visible material categories, weighs incoming and separated fractions using load cells, and creates a QR-linked, hash-chained batch record. A compatibility-first matching engine preserves individual batch identities while pooling suitable same-grade lots and ranking potential buyers by delivered cost per qualified tonne. Registered recyclers provide downstream processing and laboratory feedback.

3. Final Scope Decision

IN SCOPE NOW	FUTURE EXTENSIONS / OUT OF SCOPE
Primary demonstration: construction and demolition rubble → screened recycled aggregate → registered recycler → suitable precast paving-block manufacturer. Benchtop optical-gravimetric screening, digital passport and buyer matching.	Future matching extensions: fly ash, metallurgical slag and agricultural ash. Not current prototype outputs. No acoustic sorting, automatic chemical certification, M30+ strength guarantee, EPR escrow or zero-landfill claim.

Internal working report | Revision date: 31 October 2026 | Prepared for team alignment, technical planning and jury preparation.

HED PILLAR: QUESTION

4. Problem Context, Evidence & Regulatory Drivers

Why verified supply and material traceability matter for C&D waste exchange.

4.1 National and Local Context

India's annual C&D waste generation is commonly cited at approximately 150 million tonnes. CSE's 2026 report, Debris to Dividend, describes 34 operational C&D waste processing plants with a combined processing capacity of roughly 16,000 tonnes per day as of 2024. CSE compares this capacity with the official annual generation estimate and characterises the resulting national recovery level as approximately 4%. This figure is processing capacity-based context, not a measurement of the fraction of all waste that is actually recovered in every location. [1]

Local context: Greater Chennai Corporation removed 3.51 lakh metric tonnes of C&D waste between 7 January and 12 December 2025. The reported quantity is debris removed by the Corporation, not the total amount generated in Chennai. The material was sent to processing facilities at Kodungaiyur and Perungudi for segregation and recycling. [2]

> **WHY THE LOCAL EXAMPLE MATTERS**
>
> The Chennai figure shows that C&D debris management is an active municipal operating issue in a city relevant to the team. It does not establish that WasteWise has a Chennai pilot site or municipal partnership; those remain prospective validation steps.

4.2 Environment (C&D) Waste Management Rules, 2025

The Environment (Construction and Demolition) Waste Management Rules, 2025 were notified through G.S.R. 219(E) and came into force on 1 April 2026, superseding the 2016 framework. The rules create separate obligations and targets for EPR recycling and use of processed C&D waste in construction/reconstruction activities. [3]

FIRST SCHEDULE – EPR RECYCLING TARGETS	TARGET
2025–26	25% of recyclable C&D waste assessed for the relevant prior-year period
2026–27	50%
2027–28	75%
2028–29 onwards	100%

SECOND SCHEDULE – BUILDING / RECONSTRUCTION UTILISATION	TARGET
2026–27	5%
2027–28	10%
2028–29	15%
2029–30	20%
2030–31 onwards	25%

The Second Schedule targets refer to the percentage of total construction material required for applicable construction and reconstruction activities. The rules identify the relevant producer category by built-up area of 20,000 m^2 and above. Road-construction utilisation is addressed separately in the Third Schedule. These schedules must not be combined into one target series. [3]

4.3 Relevance to the Proposed Hub

- The system is designed to support traceable material handling and pre-dispatch evidence; it does not itself discharge a generator's legal duties.
- CPCB's C&D portal covers stakeholder registration and reporting, EPR certificates and utilisation tracking. Processed waste used to settle utilisation targets must be procured from registered recyclers. [4]

- Any proposed shared station must operate through an appropriately authorised collection point or registered intermediate waste storage facility, as applicable. The project must validate the exact operating model with the relevant local authority / CPCB process.

Sources: [1] CSE, Debris to Dividend (2026). [2] DT Next, 13 Dec 2025. [3] G.S.R. 219(E), Environment (C&D) Waste Management Rules, 2025. [4] CPCB Online C&D Waste Management System.

HED PILLAR: QUESTION

5. Root-Cause Diagnosis & Stakeholder Need

WasteWise is designed around the pre-dispatch trust gap, not only digital discovery.

5.1 Four Bottlenecks

BOTTLENECK	WHAT HAPPENS TODAY	WHAT THE SYSTEM ADDS
Quality uncertainty	Supplier photographs do not show the full distribution of wood, plastic, metal, soil or other visible contaminants in a load.	Controlled presentation, visual category estimate, physical separation and recorded weights.
Fragmented supply	Small demolition batches can fall below economical minimum loads for a recycler or buyer.	Compatibility-first consolidation of batches from multiple nearby sources.
Opaque delivered cost	Freight, hub handling, rejects, crushing, testing and final delivery are spread over the usable output.	Two-stage cost calculation per screened tonne and per qualified tonne.
Acceptance uncertainty	A buyer needs traceable information and downstream test results before treating material as a reliable input.	Tier 1 hub record plus Tier 2 recycler acknowledgement and laboratory references.

5.2 Stakeholders and Their Requirements

STAKEHOLDER	NEED / INCENTIVE	POTENTIAL VALUE
Small demolition contractors	A practical route to transfer source-segregated small batches.	Access to buyers and documented batch handover; no need to own screening hardware.
Registered recyclers	Predictable incoming quality, quantity and economical loads.	Pre-screened feed, pooled compatible lots and batch-level evidence.
Precast / paving-block manufacturers	Aggregate meeting specified material and finished-product requirements.	A traceable supply route with separate laboratory acceptance gates.
Municipalities / authorised operators	Responsible collection, storage, processing and records.	Potential shared-hub workflow and better material traceability.
Transporters / hub operators	Planned trips and adequate load utilisation.	Consolidated dispatch and service-fee opportunities.
Neighbourhoods / ecosystems	Reduced improper dumping and responsible resource use.	Potential environmental benefit where recycled material displaces virgin aggregate.

5.3 Competitive Landscape – Fair Positioning

The project background identifies Cyrkl, Excess Materials Exchange (EME), Sfridoo, Orbio, WasteTrade and Rheaply as examples of digital material discovery, reuse or exchange platforms. This report does not claim that these services lack verification, quality control or logistics features. A full competitor feature audit has not been completed.

> **WASTEWISE DIFFERENTIATION**
>
> The proposed differentiator is the integrated workflow: physical pre-dispatch screening at a shared hub, measured mass yield, traceable Tier 1 / Tier 2 batch records, compatibility-first same-grade pooling and delivered-cost ranking. It is positioned to complement existing platforms, recyclers and municipal systems.

5.4 The Core Product Hypothesis

If buyers can see measured hub screening results, preserve source-level batch identities, receive compatible pooled loads and compare cost per qualified output, they may be more willing to accept secondary feedstock. This is a hypothesis to validate through interviews, sample trials and quotations during Round 2; it is not yet a proven market result.

HED PILLAR: REIMAGINE

6. Cyber-Physical System Architecture

A shared cluster screening station serves nearby source sites and prepares traceable material lots before longer-distance dispatch.

6.1 End-to-End Workflow

01 C&D SITES	02 SHARED HUB	03 QR PASSPORT	04 AI MATCHER	05 RECYCLER
Source-segregated lots Short haul to hub	Sieve • feeder • RGB Load cells • servo gate	Batch identity Hash-chained record	Compatibility first Pooling • cost rank	Crush • grade • test Paver supply

Sequence: source segregation → screening → batch passport → compatibility-first pooling → registered recycler → suitable paving products.

> **CLOSED-LOOP FEEDBACK**
>
> The recycler's confirmed received quantity, processing yield, aggregate grading and laboratory report references are added to Tier 2 of the batch passport and countersigned by the recycler.

6.2 Layer 1 – Mechatronics: Screening Station

SUBSYSTEM	FUNCTION / DESIGN INTENT
Inclined mechanical sieve	Separates finer mineral particles from coarse rubble using a selected mesh aperture. Fines are collected and weighed separately; their reuse is conditional on suitability checks.
Single-layer vibratory / belt feeder	Presents coarse material in a controlled, non-overlapping layer so the camera can observe individual items more consistently and the diverter can act sequentially.
Controlled-LED RGB hood	Provides consistent illumination and reduces variation from outdoor shadows and glare during image capture.
HX711 load cells + ESP32	Records incoming and separated-bin mass. The ESP32 coordinates sensor readings and the servo mechanism; load-cell calibration is required.
Servo-operated binary diverter	Diverts visible macro-rejects into a reject bin while passing the accepted mineral stream. It does not independently certify chemical or structural suitability.
Collection bins	Keep accepted coarse mineral, fines and visible rejects separate for weighing, record-keeping and downstream disposition.

6.3 Prototype Scale Boundary

Round 2 is a benchtop proof-of-concept that validates synchronized presentation, visual classification, weighing, diversion and passport logic. A commercial shared hub would require a wider belt, higher-throughput separation, dust management, safe material handling and site-specific engineering. Pneumatic air-jets or manual pick stations are scale-up options, not claimed prototype capabilities.

HED PILLAR: DEVELOP

7. AI/ML Layer, Digital Material Passport & Batch Pooling

The passport distinguishes measured hub data from recycler-confirmed downstream results.

7.1 Three-Class RGB Vision

The vision model is scoped to three visual classes: (1) concrete, (2) brick / masonry and (3) visible rejects. The model provides a preliminary visual screening estimate for the material passing under controlled illumination. The servo gate performs a binary physical decision: accepted mineral stream versus visible reject stream.

IMPORTANT MODEL LIMITATION

RGB surface images do not reliably establish the mass ratio of concrete to brick within a batch, hidden inclusions, chemical contaminants or compressive strength. Do not present pixel-based visual classification as a strength grade or material certification.

7.2 Tiered Digital Material Passport

PASSPORT FIELD	TIER 1 – HUB RECORD	TIER 2 – RECYCLER RECORD
Identity	Unique batch UUID, source / generator reference, hub ID, timestamp and inspection record.	Received shipment reference and recycler identity.
Measured mass	Incoming, accepted mineral, fines and reject weights; calculated Y_screen.	Received mass and post-processing output mass; calculated Y_proc.
Visual / process data	Sieve aperture, visual class estimate, inspection photographs and operator review flags.	Processing method / lot reference and aggregate grading data, where supplied.
Integrity / sign-off	SHA-256 hash-chained record and hub digital signature using a protected key.	Recycler countersignature appended to Tier 2 acknowledgement.
Laboratory / acceptance	No statutory quality certificate implied by the hub record.	Laboratory report references, buyer acceptance / rejection and reason codes where available.

The QR code points to the batch record; the QR image itself is not the proof of integrity. Integrity is supported by the stored hash chain and the hub signature. The proposed signing approach requires secure private-key storage, access controls and a verifiable signature process during implementation.

7.3 Compatibility-First Matching

1. Hard compatibility gate: compare material type, sieve aperture / gradation, visual grade band, reject threshold and the buyer's stated acceptance requirements before pooling.
2. Same-grade pooling: consolidate only batches that pass the compatibility gate. Do not combine unrelated waste streams merely to fill a truck.
3. Sub-batch traceability: each contributing batch retains its own QR identity and remains linked to the consolidated shipment manifest.
4. Delivered-cost ranking: rank eligible buyers using transport, hub handling, processing, testing and final delivery costs per qualified tonne.
5. Recycler feedback: use verified Tier 2 yield and laboratory status to refine future acceptance rules without relabelling training images based on unrelated commercial rejection reasons.

7.4 Yield Variables

VARIABLE	MEANING	SOURCE
Y_screen = M_accepted / M_incoming	Hub screening yield	Calibrated load-cell measurements.
Y_proc = M_qualified / M_accepted	Recycler processing recovery	Recycler-confirmed mass after processing and qualification.
Y_overall = Y_screen × Y_proc	Illustrative end-to-end yield	Combined hub and recycler measurements.
C_screened	Cost per screened tonne	Stage 1 costs / screened accepted mineral mass.
C_qualified	Cost per qualified tonne	End-to-end costs / qualified aggregate output.

HED PILLAR: REIMAGINE

8. Circular Construction Pathway, Quality Gates & Residual Accounting

The current demonstration is construction rubble to screened aggregate to suitable paving products.

8.1 Primary Hero Pathway

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5

Hub screen Sieve, inspect, weigh	**Recycler** Crush / process	**Aggregate gate** Applicable IS 383 tests	**Precast plant** Paving-block manufacture	**Product gate** Applicable IS 15658 tests

Primary output: screened coarse mineral material moves to a registered C&D recycler for crushing and grading. Suitable recycled aggregate may then be used by a precast manufacturer after the relevant aggregate and finished-product acceptance tests.

8.2 Standards as Downstream Testing Targets

- IS 383:2016 – Coarse and Fine Aggregate for Concrete (Third Revision), including amendments and current applicability, is a target reference for aggregate testing. BIS lists the standard as reaffirmed in July 2025. [5]
- IS 15658:2021 – Concrete Paving Blocks (First Revision), including applicable amendments, is a target reference for finished paving-block testing. BIS implementation material identifies Amendment No. 1. [6]
- Passing hub screening does not establish compliance with either standard. The receiving recycler, concrete mix designer and accredited laboratory determine applicable test methods, acceptance criteria and suitability.

8.3 Honest Fraction-by-Fraction Accounting

FRACTION	CURRENT HANDLING	CLAIM BOUNDARY
Coarse mineral accepted (M_accepted)	Sent to registered recycler for crushing / processing, subject to receipt and acceptance.	Do not call it qualified aggregate until downstream processing and tests.
Fines (M_fines)	Kept separate, weighed and assessed for potential application.	Do not automatically count as reused; suitability must be confirmed.
Visible rejects (M_reject)	Separated and recorded. Potential recovery routes may be explored where suitable and authorised.	Unrecoverable residue must be accounted for and costed for lawful disposal.
Recycler processing loss	Recorded as the difference between accepted feed and qualified output.	Do not combine this with the hub screening rejects or fines.

> **ZERO-LANDFILL POSITION**
>
> The challenge brief includes a closed-loop / zero-landfill ambition. WasteWise Verify supports higher recovery and more honest residual accounting, but this project does not guarantee zero landfill. Every fraction needs a verified destination; residual disposal remains visible in the model.

8.4 Future Industrial-Symbiosis Extensions

The matching architecture could later connect compatible byproducts such as fly ash, metallurgical slag and agricultural ash with suitable manufacturers. Additional future streams may include rice-husk ash, textile waste or other industrial byproducts. These are post-Round 2 extensions. Each stream requires separate sampling, chemical and physical compatibility checks, process design and finished-product performance validation. No co-blending recipe or strength grade is promised in the current project.

HED PILLAR: IMPACT

9. Two-Stage Yield & Delivered-Cost Model

A reconciled hypothetical example compares direct supply with a pooled shared-hub route.

9.1 Physical Assumptions – 10.0 Tonne Batch

INPUT / OUTPUT	ASSUMPTION	CALCULATION
Incoming construction rubble	10.0 t	Three nearby sources: 3.0 t + 4.0 t + 3.0 t.
Coarse mineral accepted	8.0 t	Y_screen = 8.0 / 10.0 = 0.80.
Separated fines	0.8 t	Recorded separately; not automatically counted as reused.
Visible rejects	1.2 t	Costed for disposal at ₹1,000/t in both scenarios.
Recycler processing recovery	87.5%	Y_proc = 7.0 / 8.0 = 0.875.
Qualified aggregate output	7.0 t	Y_overall = 0.80 × 0.875 = 0.70.

9.2 Two-Stage Cost Equations

STAGE 1 – COST PER SCREENED TONNE

C_screened = (P + T1 + H + D + T2) / (M_incoming × Y_screen).
The denominator is the 8.0 t accepted coarse mineral stream in this example.

STAGE 2 – COST PER QUALIFIED TONNE

C_qualified = (P + T1 + H + D + T2 + R + Q + T3) / (M_incoming × Y_screen × Y_proc).
The denominator is 7.0 t of assumed qualified aggregate output.

Variables: P = raw material procurement / acquisition; T1 = site-to-hub transport; H = hub screening fee; D = legal residual disposal; T2 = hub-to-recycler transport; R = recycler sorting and crushing; Q = aggregate laboratory testing; T3 = delivery to the paving-block manufacturer.

9.3 Reconciled Illustrative Cost Ledger

COST ITEM	DIRECT SUPPLY	WASTEWISE HUB	BASIS
P – Material procurement	₹1,000	₹1,000	Identical acquisition cost on 10.0 t.
T1 – Site-to-hub local haulage	₹0	₹2,400	Three short local drops at ₹800 each.
H – Shared hub screening fee	₹0	₹2,000	₹200/t on 10.0 t intake.
D – Legal reject disposal	₹1,200	₹1,200	1.2 t at ₹1,000/t in both cases.
T2 – Haulage to recycler	₹12,000	₹4,600	Three under-loaded trips vs. one consolidated load.
STAGE 1 SUBTOTAL	₹14,200	₹11,200	Direct: ₹1,775 / accepted t; hub: ₹1,400 / accepted t.
R – Recycler sorting & crushing	₹6,500	₹3,200	Dirty mixed feed vs. pre-screened feed assumption.
Q – Aggregate laboratory testing	₹2,400	₹2,400	Same hypothetical batch test cost.
T3 – Delivery to paver plant	₹2,100	₹2,100	Same delivery cost for 7.0 t output.
END-TO-END TOTAL	₹25,200	₹18,900	Same assumed 7.0 t qualified output.
C_qualified	₹3,600/t	₹2,700/t	25.0% illustrative reduction; ₹900/t saved.

9.4 Why the Example Is Transparent

Both columns retain the same acquisition cost, legal reject disposal cost, laboratory test cost, final delivery cost and 7.0 t qualified output. In the hypothetical hub route, the model changes local transport, consolidated transport, hub screening and recycler sorting / crushing assumptions. The 25% difference is generated by these stated assumptions; it is not a measured result or guaranteed saving.

ROUND 2 VALIDATION REQUIRED

Replace every assumed transport, hub, crusher and testing rate with written local quotations or measured pilot data. Reconcile costs over total recovered mass for pooled lots: total batch costs divided by total qualified recovered mass.

HED PILLAR: DEVELOP

10. Hub Governance, Commercial Model & Team Plan

A shared service model avoids asking every small contractor to buy dedicated screening hardware.

10.1 Proposed Hub Operating Model

A shared cluster screening station is proposed at an appropriately authorised C&D collection point, registered intermediate storage facility, municipal transfer yard or registered recycler spoke, depending on local permissions and operating arrangements. The hub serves nearby source sites within a catchment area that must be selected

through field validation. The platform provides the screening record, QR passport, compatibility filter, pooling logic and delivered-cost comparison.

10.2 Three Revenue Streams

REVENUE STREAM	PAYER / BASIS	VALUE OFFER
1. Generator intake / screening fee	Contractor or generator; per batch or per tonne.	Screening, measured batch record and traceable handover.
2. Consolidated dispatch fee	Buyer / recycler; per matched shipment or dispatch.	Compatible pooled loads and reduced under-loaded freight risk.
3. Hub SaaS / equipment lease	Hub operator; subscription or lease.	Passport hosting, batch tracking, pooling and delivered-cost tools.

These are proposed revenue streams for validation through industry interviews. Willingness to pay, pricing, capital expenditure, operating expenditure and the hub operator's responsibilities remain to be established.

10.3 Governance and Compliance Boundaries

- WasteWise is an operational technology layer; it is not a statutory C&D waste compliance portal or EPR certificate issuer.
- Appropriate registration and authorisation for collection points, intermediate storage facilities, recyclers and other relevant entities must be confirmed for the pilot structure.
- CPCB's portal provides registration, reporting, EPR certificate and processed-waste utilisation functions. WasteWise batch records can support operational traceability but do not replace those functions. [4]
- Material used to meet regulated utilisation targets must follow the applicable rules and be sourced from registered recyclers as specified by the portal / rule framework.

10.4 Five-Member Responsibility Matrix

ROLE	PRIMARY RESPONSIBILITY	KEY DELIVERABLE
Team Lead – Mechatronics	System integration, mass-balance logic, economics, regulatory research and final deck consistency.	Integrated plan, cost model and master presentation.
Mechatronics – Mechanical / Embedded	Sieve, feeder, LED hood, load-cell mounting, ESP32 and servo gate.	CAD, wiring and functioning benchtop rig.
AI/ML 1 – Computer Vision	Controlled RGB dataset and 3-class concrete / brick-masonry / visible-reject classifier.	Dataset, training pipeline and evaluation.
AI/ML 2 – Matching & Cost	Rule-first compatibility, same-grade pooling, route comparison and cost per qualified tonne.	Matching logic, pooling simulation and cost ranker.
AI/ML 3 – Passport & Web	Batch ID, QR record, hash chain, hub signature, Tier 2 recycler interface and dashboard.	Digital passport prototype and web interface.

10.5 Benchtop Prototype Bill of Materials – Indicative

ITEM	ROLE / NOTES
Inclined wire-mesh sieve frame	Interchangeable mesh aperture for coarse / fines separation.
DC vibratory or belt feeder	Controlled single-layer presentation; guard moving parts.
Diffused 5000K LED enclosure	Consistent RGB image capture; dust-managed hood.
RGB camera + laptop / Raspberry Pi inference	Image capture and 3-class model execution; select platform based on latency and cost.
ESP32 microcontroller	Sensor / actuator coordination and data exchange.
HX711 + load-cell assemblies	Measure incoming and separated material mass; calibrate against known weights.
Metal-gear servo diverter	Binary accepted-mineral / visible-reject diversion.
Collection bins and frame	Separate coarse accepted mineral, fines and visible rejects.

The BOM is a proposed Round 2 specification, not a purchase record or confirmed prototype inventory. Component ratings, safety guards, dust protection and capacity must be finalised during CAD and sourcing.

HED PILLAR: DEVELOP

11. Round 2 Execution Plan, Validation & Risk Controls

A realistic benchtop proof-of-concept is the next milestone; the current concept is not yet a validated industrial plant.

11.1 Eight-Week Development Roadmap

PHASE	WORK PACKAGES	OUTPUT / EXIT CHECK
Weeks 1–2 | CAD & preparation	Finalise sieve / feeder CAD, LED enclosure, mechanical flow, wiring plan, cost model and pooling simulator. Confirm a suitable source of representative rubble samples and component availability.	Reviewed CAD, component list, sample plan and initial cost / pooling model.
Weeks 3–4 | Hardware build	Assemble feeder, load cells, HX711, ESP32, RGB camera, LED hood, servo gate and separate bins. Begin controlled image capture.	Benchtop rig moves and separates material; sensor data can be logged.
Weeks 5–6 | Calibration & integration	Calibrate load cells using known weights; conduct manual-sort comparisons; run representative oven-dry moisture checks; test mass balance; implement QR passport and pooling demonstration.	Repeatable weight readings, documented classification test and functioning Tier 1 passport.
Weeks 7–8 | Validation & user feedback	Run repeated sample trials; request recycler / transport quotations; interview at least three recyclers, contractors or paver manufacturers; seek a potential authorised hub operator; prepare demo.	Validation log, interview findings, revised cost model and working demonstration.

11.2 Validation Matrix

TEST	METHOD	WHAT IT ESTABLISHES
Load-cell calibration	Use known reference weights at several points and repeat readings.	Measurement error, repeatability and tare stability.
Mass-balance closure	Weigh incoming mass and all separated output bins.	Whether recorded fractions account for the batch within a stated tolerance.
Vision comparison	Compare model predictions against manually labelled images / items.	Class-wise precision, recall, confusion and failure cases; report measured values only.
Moisture check	Oven-dry representative samples from relevant fractions.	Moisture differences that may affect interpretation of recorded wet mass.
Gate operation	Run repeated accepted / visible-reject samples and count successful diversions.	Actuator timing, jams, missed rejects and safe fallback behaviour.
Passport integrity	Generate, store, retrieve and verify hash-chained records and signatures.	Record consistency, identity preservation and Tier 2 acknowledgement workflow.
Buyer matching	Use test buyer specifications and sample batch data.	Whether compatibility filtering occurs before pooling and cost ranking.
Economics validation	Replace modelled rates with local written quotations / measured costs.	Whether the modelled cost advantage persists under local conditions.

11.3 Key Risks and Mitigations

RISK	MITIGATION / HONEST BOUNDARY
Dust, overlap and irregular shapes	Controlled LED hood, singulation, labelled dataset and manual-review fallback. Vision remains a surface-level indicator.
Hidden chemical contaminants	Not detectable by RGB vision; retain downstream sampling and laboratory gates.
Moisture variation	Record wet mass and test representative samples by oven-dry method; do not assume all fractions have identical moisture.
Sensor drift / vibration	Stable mounting, tare procedures, known-weight calibration and repeatability checks.
Unreliable buyer / market assumptions	Conduct interviews and replace hypothetical rates with quotations; do not claim guaranteed savings.
Regulatory uncertainty at a site	Select an appropriately authorised / registered operating partner and confirm local permissions before pilot operation.

RISK	MITIGATION / HONEST BOUNDARY
Unusable fines or rejects	Keep fractions separate; assess potential outlets; include lawful residual disposal in the ledger.

ROUND 1 PRESENTATION ALIGNMENT

12. Six-Slide Deck Blueprint & Jury Defence

This report supports the revised six-slide official RE:IMAGINE presentation.

SLIDE	PURPOSE	CORE MESSAGE / VISUAL
1. Project dossier	Introduce WasteWise Verify and the team.	Title, tagline, secondary-feedstock subtitle, RE:MAKE / HED8-MAK-04 and team roster.
2. Investigation	Show the C&D problem, stakeholders, root causes and white-space.	Updated CSE / Chennai context, separate regulatory schedules, four bottlenecks and fair competitor positioning.
3. Solution design	Explain the cyber-physical workflow.	Five-node horizontal architecture, compatibility-first matcher, Tier 2 feedback, yield / mass-balance formula and camera boundary.
4. Multidisciplinary pathways	Show how Mechatronics, AI/ML, design and policy connect.	Three columns: hardware / software, operator-centred design and commercial / regulatory model; future symbiosis extension.
5. Feasibility & impact	Show the assumed mass balance, economic potential and Round 2 plan.	10 t → 8 t + 0.8 t + 1.2 t; 8 t × 87.5% = 7 t; testing plan and eight-week roadmap.
6. Closing	Close with official wording and required submission details.	Preserve official closing statement and four equally weighted jury criteria; use real details only as allowed by blind-review instructions.

12.1 Jury Questions to Prepare

Q1. Why not let the recycler sort material at its own plant?

The shared hub is intended to screen near source, measure usable mineral mass and consolidate compatible small lots before longer-haul transport. The model must validate whether the added hub stop is offset by better load utilisation and lower downstream sorting costs.

Q2. Can the camera detect gypsum, sulphates or chlorides?

No. RGB vision identifies visible surface categories and macro-rejects only. Hidden or chemical contaminants require appropriate sampling and laboratory testing.

Q3. Why use three vision classes with a binary gate?

The model labels concrete, brick / masonry and visible rejects for descriptive screening. The physical gate makes a simpler binary separation: accepted mineral stream versus visible reject. Vision does not certify composition by mass or strength.

Q4. How can a benchtop rig represent an industrial hub?

It cannot reproduce full industrial throughput. Round 2 validates synchronized presentation, imaging, weighing, diversion and data logic at benchtop scale. Wider conveyors and industrial separation equipment require later engineering.

Q5. Why would a small contractor pay a screening fee?

The value hypothesis is that a contractor may gain an organised handover, traceable batch information and access to pooled loads. Willingness to pay must be tested with contractors and hub operators.

Q6. Is the 25% cost reduction guaranteed?

No. The 10-tonne comparison is hypothetical. It keeps the same qualified output and major common costs, while using assumptions for freight, hub and processing costs. Round 2 must replace those assumptions with local quotations or measured pilot data.

Q7. Does the QR passport provide CPCB compliance or an EPR certificate?

No. It is an operational traceability record. Registration, statutory reporting, EPR certificates and utilisation settlement remain within the applicable CPCB / regulatory framework.

Q8. What happens to fines and rejects?

They are weighed and recorded separately. Fines are not counted as reused until their suitability and destination are confirmed. Rejects are routed only where a valid recovery outlet exists; residual disposal remains costed.

12.2 Unified 30-Second Team Pitch

> **ELEVATOR PITCH**
>
> Construction waste is often difficult to reuse because buyers cannot confidently assess small, mixed batches before paying to transport them. WasteWise Verify proposes a shared screening hub that separates and weighs construction rubble, records each batch in a traceable QR passport, and uses compatibility-first matching to pool suitable lots and rank buyers by delivered cost per qualified tonne. Registered recyclers then process and test the material before it reaches paving-block manufacturers.

REFERENCE & RELEASE CONTROL

13. Sources, Assumptions & Final Team Audit

Use this page to maintain a defensible, internally consistent master version.

13.1 External Sources

[1] Centre for Science and Environment (CSE). Debris to Dividend (2026). Used for the 34 operational plants, approximately 16,000 tonnes/day capacity as of 2024, and comparison with the official 150 million tonne annual estimate.
https://www.cseindia.org/content/downloadreports/13045

[2] DT Next. "GCC wrapped '25: 3.51 lakh MT of construction debris removed in Chennai." Updated 13 December 2025. The figure refers to debris removed between 7 January and 12 December 2025.
https://www.dtnext.in/amp/news/chennai/gcc-wrapped-25-351-lakh-mt-of-construction-debris-removed-in-chennai-856548

[3] Government of India, Ministry of Environment, Forest and Climate Change. G.S.R. 219(E), Environment (Construction and Demolition) Waste Management Rules, 2025. Notified 2 April 2025; effective 1 April 2026. Includes First, Second and Third Schedules.
https://egazette.gov.in/WriteReadData/2025/262313.pdf

[4] Central Pollution Control Board (CPCB). Online C&D Waste Management System – registration, reporting, EPR certificate and utilisation functions.
https://cdwm.cpcb.gov.in/about-system

[5] Bureau of Indian Standards (BIS). IS 383:2016, Coarse and Fine Aggregate for Concrete – Specification (Third Revision). BIS standard details indicate review / reaffirmation in 2025.
https://standards.bis.gov.in/website/standard-details

[6] Bureau of Indian Standards (BIS). IS 15658:2021, Concrete Paving Blocks – Specification (First Revision). Refer to the applicable amendment and current testing requirements.
https://www.bis.gov.in/wp-content/uploads/2022/06/PM-IS-15658-June-2022.pdf

13.2 Assumptions and Claims Register

ITEM	STATUS / HOW TO PRESENT
150 million tonnes/year	Officially cited estimate used as national context; CSE describes it as an official estimate and discusses uncertainty.
16,000 tonnes/day / 34 plants	CSE-reported combined processing capacity as of 2024; not a direct measure of actual recovered tonnage.
3.51 lakh tonnes Chennai	Municipal removal quantity during the stated 2025 period; not city-wide generation.
3–4 tonne source lots	Illustrative planning assumption; validate with local generators / recyclers.
10 tonne mass balance	Hypothetical: 8.0 t coarse + 0.8 t fines + 1.2 t rejects; 7.0 t qualified output assumes 87.5% recycler recovery.
25% cost reduction	Modelled from stated cost assumptions; not a measured or guaranteed saving.
Environmental benefit	To be measured in Round 2 using published factors and transport / electricity data.
Future byproduct matching	Concept extension only; material compatibility and product performance require separate validation.
Prototype and pilot partner	Proposed for Round 2; do not state that the rig has been built or a partner secured unless verified.

13.3 Final Release Checklist

- Use WasteWise Verify as the sole project brand across all materials.
- Keep the primary pathway to construction rubble → screened aggregate → registered recycler → paving blocks.

- Keep fly ash, slag and agricultural ash as future extensions only.
- Use Y_screen, Y_proc, Y_overall, C_screened and C_qualified consistently.
- Keep hub screening mass balance separate from recycler processing recovery.
- Keep fines and rejects separately accounted; do not claim automatic reuse or zero landfill.
- Label all illustrative figures and modelled savings clearly.
- Confirm actual hub / collection-point registration and local permissions before any pilot.
- Use real team details only where the submission rules permit; verify blind-review requirements.
- Before jury presentation, ensure every member can explain the complete workflow and limitations.

13.4 Quick Technical Reference

TERM / DECISION GATE	WORKING DEFINITION
Y_screen	Hub screening yield = M_accepted / M_incoming; based on calibrated load-cell mass.
Y_proc	Recycler processing recovery = M_qualified / M_accepted; confirmed after processing.
C_screened	Stage 1 cost per accepted screened tonne, including acquisition, T1, hub, disposal and T2 costs.
C_qualified	End-to-end cost per qualified tonne after recycler processing, testing and final delivery.
Tier 1	Hub-origin record: identity, timestamp, weights, sieve / visual data, hash chain and hub signature.
Tier 2	Recycler acknowledgement: received quantity, processing yield, grading / lab references and countersignature.
Compatibility-first pooling	A batch must pass buyer material / quality rules before being combined with other lots.
Moisture validation	Oven-dry checks on representative samples during Round 2; no dedicated moisture sensor in current BOM.

CLAIMS TO AVOID

Do not claim that the camera certifies chemical quality or strength, that fines are automatically reused, that the current concept guarantees zero landfill, that the 25% saving is measured, or that the benchtop rig / pilot partnership already exists.

CORE MESSAGE TO THE JURY

WasteWise Verify establishes what is physically usable, preserves its batch identity, and connects compatible materials to suitable industries before long-distance dispatch.

Prepared from the previous WasteWise Verify Project Brief & Execution Plan, the revised six-slide proposal and the team's finalised scope decisions. External references checked for this revision on 31 October 2026.